

Bipartitions and Beatty Sequences

Spring 2021 ARML Power Contest

The Background

A *partition* of a set is simply a way of splitting the set into nonempty subsets, called *parts*, that don't overlap and that fill up the whole set. For example, a partition of $\{1, 2, 3, 4, 5\}$ could be $\{1, 3\}, \{2, 5\}, \{4\}$. A *bipartition* is specifically a partition into two parts. Given a subset A of a set U , the set B of all elements of U not belonging to A is called the *complement* of A , denoted A' . So a subset A of U together with its complement A' form a bipartition of U , provided neither A nor its complement A' is empty. This contest will investigate some bipartitions of the positive integers.

This contest frequently uses the floor and ceiling functions, $\lfloor x \rfloor$ and $\lceil x \rceil$ respectively. The floor function gives the greatest integer less than or equal to x while the ceiling gives the least integer greater than or equal to x . The following facts about the floor function may be used without further proof:

- For any real number x , $\lfloor x \rfloor \leq x < \lfloor x \rfloor + 1$.
- For any real number x and any integer n , $\lfloor x + n \rfloor = \lfloor x \rfloor + n$.

The Problems

There are 40 points available on this contest. The numbers at the end of each problem are the number of points that problem is worth.

1. Given that the set of odd positive integers, $\{1, 3, 5, 7, \dots\}$ is one set in a bipartition of the positive integers, determine its complement. [1 pt.]
2. The set of prime positive numbers and the set of composite positive numbers taken together *do not* form a bipartition of the positive integers. Why not? [1 pt.]

Let $x > 1$ be a rational number. Consider the infinite sequence

$$X = \{\lfloor x \rfloor, \lfloor 2x \rfloor, \lfloor 3x \rfloor, \dots, \lfloor kx \rfloor, \dots\}$$

for all positive integers k . For example, if $x = \frac{7}{5}$ then $X = \{1, 2, 4, 5, 7, 8, 9, 11, 12, 14, \dots\}$. Notice that this is not the set of all positive integers. But there are definitely patterns in the numbers that are included in X , and in those that are in its complement X' .

3. Let $x = \frac{p}{q}$ in lowest terms, so that p and q are positive integers with no common factors.
 - (a) Given that positive integer n is in X , show that $n + p$ is also in X . [2 pts.]
 - (b) Given that positive integer m is in X' , show that $m + p$ is also in X' . [2 pts.]
4. If $p > 2q$ then the least number in X is $\lfloor \frac{p}{q} \rfloor \geq 2$ and the least number in X' is clearly 1. On the other hand, if $q < p < 2q$, show that the least positive integer in X' is $\lceil \frac{q}{p-q} \rceil$. [2 pts.]

The consequence of problem 3 is that X consists of a union of arithmetic sequences whose common differences are each equal to p , while the complementary set of all positive integers not in X is likewise a union of arithmetic sequences with common difference p . So the bipartitions of the positive integers created by looking at the whole part of multiples of a rational number are very predictable and boring. Not so for *irrational* numbers!

For an irrational number x , the sequence $\{\lfloor x \rfloor, \lfloor 2x \rfloor, \lfloor 3x \rfloor, \dots\} = \{\lfloor kx \rfloor\}_{k=1}^{\infty}$ is called the *Beatty sequence* generated by x .

5. Compute the first ten terms of the Beatty sequence generated by $\phi = \frac{1+\sqrt{5}}{2} \approx 1.618$. [2 pts.]

6. Compute the first ten terms of the Beatty sequence generated by $\phi^2 = \frac{3+\sqrt{5}}{2} \approx 2.618$. [2 pts.]

It appears as if the sequence for ϕ^2 fills in the gaps of the sequence for ϕ and *vice versa* (hint: if this isn't the case for your answers to the previous two problems, go back and check your arithmetic!). Could it be that these two Beatty sequences are actually the complementary sets of a bipartition?

First, note that $\phi^2 = \phi + 1$, so ϕ is the positive solution to the polynomial equation $x^2 - x - 1 = 0$.

7. Show that $\frac{1}{\phi} + \frac{1}{\phi^2} = 1$. [1 pt.]

8. Let α and β be any two irrational numbers greater than 1 such that $\frac{1}{\alpha} + \frac{1}{\beta} = 1$. Prove that there are no positive integers m , i , and j such that $m = \lfloor i\alpha \rfloor = \lfloor j\beta \rfloor$. That is, with the given restrictions on α and β , no number appears in both of their Beatty sequences. [3 pts.]

9. Under the same conditions on α and β , as in the previous problem, prove that every positive integer appears in either the Beatty sequence generated by α or the Beatty sequence generated by β , or both. [3 pts.]

So the answer to the question asked above is “Yes!” In fact, for any two irrational numbers α and β satisfying the conditions that α and β are both greater than 1 and $\frac{1}{\alpha} + \frac{1}{\beta} = 1$, the corresponding Beatty sequences do not overlap (problem 8) but do contain every positive integer (problem 9). Problem 7 shows that ϕ and ϕ^2 satisfy the condition, so their Beatty sequences are the complementary sets of a bipartition.

10. Show that the Beatty sequence for an irrational number $\alpha > 1$ can never contain all the terms of any nonconstant arithmetic sequence. [2 pts.]
11. While the facts about complementary Beatty sequences are fascinating enough, show that the difference between the n th term of the sequence generated by ϕ^2 and the n th term of the sequence generated by ϕ is exactly n . [2 pts.]

12. The complementarity doesn't quite work for rational numbers. Let $x = \frac{p}{q}$ in lowest terms and $y = \frac{p}{p-q}$ so that x and y are both greater than 1 and $\frac{1}{x} + \frac{1}{y} = 1$. Prove that the sequences $X = \{\lfloor kx \rfloor\}_{k=1}^{\infty}$ and $Y = \{\lfloor ky \rfloor\}_{k=1}^{\infty}$ are not complementary by showing that:

- (a) $X \cap Y = \{kp\}_{k=1}^{\infty}$, the sequence of positive multiples of p . [2 pts.]
 (b) $X \cup Y$ contains all positive integers *excluding* the sequence $\{kp - 1\}_{k=1}^{\infty}$. [2 pts.]

Now let $0 < \alpha < 1$ be an irrational number. The sequence $\{\lfloor k\alpha \rfloor\}_{k=1}^{\infty}$ will be called a *slow Beatty sequence*. For example, let $\alpha = \frac{\pi}{8} \approx 0.393$. Then the first fifteen multiples of α are approximately

0.393, 0.785, 1.178, 1.571, 1.964, 2.356, 2.749, 3.142, 3.534, 3.927, 4.320, 4.712, 5.105, 5.500, 5.890,

so the slow Beatty sequence generated by $\frac{\pi}{8}$ is

$$0, 0, 1, 1, 1, 2, 2, 3, 3, 3, 4, 4, 5, 5, 5, \dots$$

Let X_α be the sequence of positions in the slow Beatty sequence generated by α where the sequence stays the same between the current term and the next, and let Y_α the sequence of positions where the slow Beatty sequence generated by α increases by 1. For $\alpha = \frac{\pi}{8}$, the above computations show that

$$\begin{aligned} X_{\pi/8} &= 1, 3, 4, 6, 8, 9, 11, 13, 14, \dots \\ Y_{\pi/8} &= 2, 5, 7, 10, 12, \dots \end{aligned}$$

A terser definition of X_α and Y_α is that an integer n is in X_α whenever $\lfloor n\alpha \rfloor = \lfloor (n+1)\alpha \rfloor$ and n is in Y_α whenever $\lfloor n\alpha \rfloor + 1 = \lfloor (n+1)\alpha \rfloor$. Since $\alpha < 1$, the slow Beatty sequence generated by α can never increase by more than 1, so X_α and Y_α form a bipartition of the positive integers.

13. Prove that Y_α is the Beatty sequence generated by $\frac{1}{\alpha}$. [4 pts.]

Of course, there are many bipartitions of the integers not involving Beatty sequences. Start with any nondecreasing sequence of nonnegative integers. As an example, try $Q = \{0, 2, 2, 2, 4, 4, 4, 4, 4, 5, 5, 6, 9, \dots\}$ where there is no implied pattern followed by Q past what is shown. For such a sequence, define its *frequency sequence* to be $Q^* = \{f_1, f_2, f_3, \dots\}$ where f_j is the number of entries of Q less than j . For example, in Q there are 4 entries less than 3, so $f_3 = 4$. Thus, $Q^* = \{1, 1, 4, 4, 9, 11, 12, 12, 12, \dots\}$ and f_{10} and beyond cannot be determined from the portion of Q shown.

14. Let Q be any nondecreasing sequence of nonnegative integers, and let Q^* be its frequency sequence. Prove that Q^{**} , the frequency sequence of the frequency sequence, is equal to the original sequence Q . That is, prove that $Q = Q^{**}$. [3 pts.]
 15. Let Q be any nondecreasing sequence of nonnegative integers, $Q = \{q_1, q_2, q_3, \dots\}$, and define the sequence $P = \{q_1 + 1, q_2 + 2, q_3 + 3, \dots\}$. Prove that P is a strictly increasing sequence of positive integers. [2 pts.]

16. Let Q and P be as defined in the previous problem, and let Q^* be the frequency sequence of Q , given by $Q^* = \{f_1, f_2, f_3, \dots\}$. Define P^* from Q^* as in the previous problem: $P^* = \{f_1 + 1, f_2 + 2, f_3 + 3, \dots\}$. Prove that P^* is, in fact, the complement P' . [4 pts.]